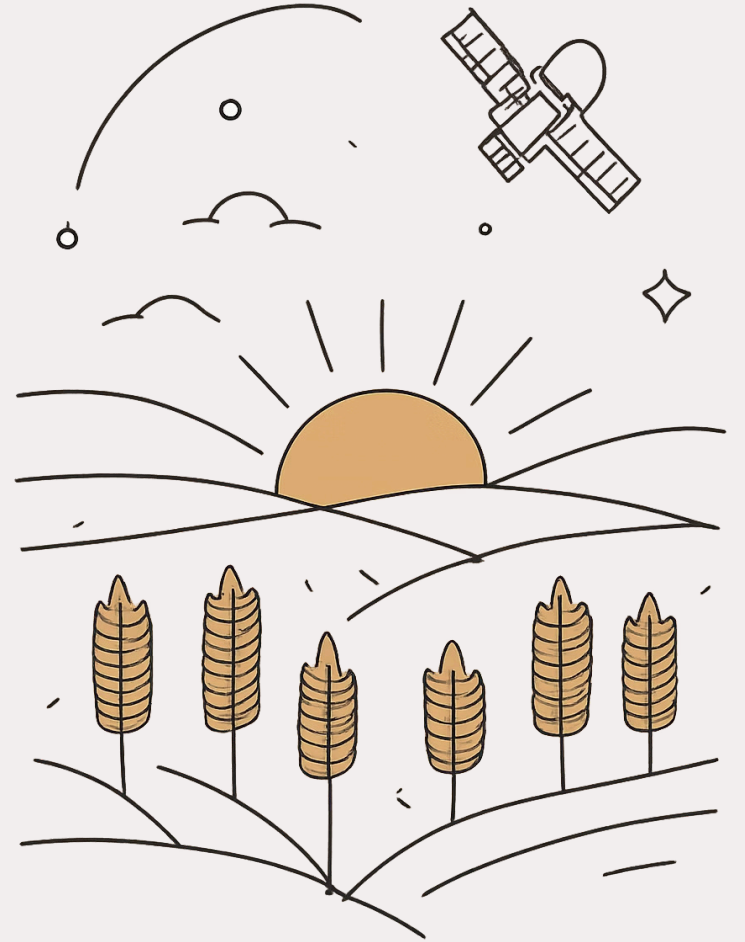


Agri-Block

Trustless Crop Trading, From Seed to Settlement

A decentralized agricultural forward-contracts platform — satellite-verified Crop NFTs, automated USDC escrow, and on-chain transparency for farmers, buyers, and validators.



Billions in Crop Value Trade on Trust Alone

Buyers Can't Verify Before Paying

No way to confirm a crop exists or is healthy before committing funds.

Farmers Wait Weeks for Payment

Manual inspections, paper records, and delayed settlements leave farmers squeezed on price.

No Recourse When Deals Fail

Paper contracts offer no automated enforcement — disputes drag on for months.

Smallholders Locked Out

Formal commodity markets remain inaccessible without expensive brokers and manual reports.



Three Technologies. One Trustless System.

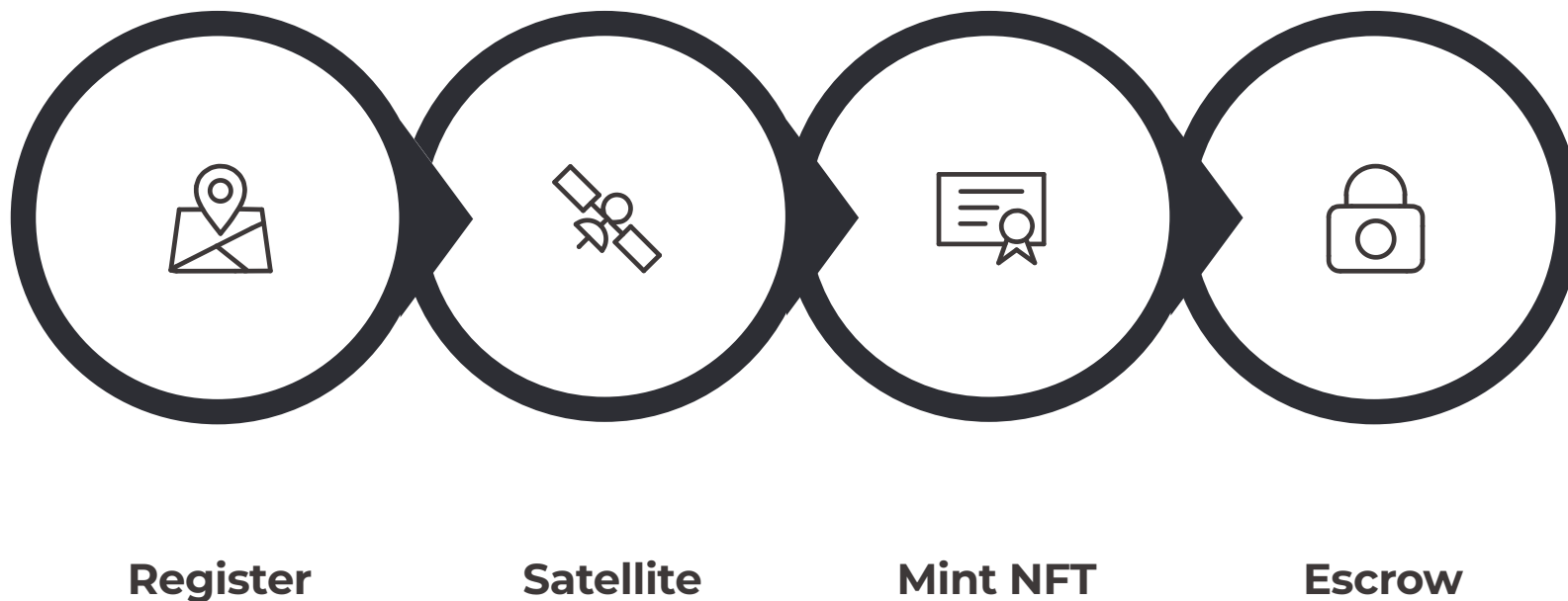
Agri-Block replaces paper contracts and manual inspections with math, code, and satellite data.

- **Satellite data** proves the crop exists and is healthy
- **Smart contracts** enforce payment automatically on verified delivery
- **AI summaries** make complex NDVI data readable for everyone

❗ Instead of trusting paper reports, both sides trust math.

HOW IT WORKS

From Parcel to Payment in Four Steps



Every step is timestamped, hashed, and recorded on Stellar — a fully auditable trail from seed to settlement.

FOR FARMERS

Working Capital Before Harvest



Register Your Land

Draw parcel boundaries on an interactive map. No broker needed.



Free Satellite Check

Copernicus Sentinel-2 scans your crop. AI explains what the data means.



20% Upfront USDC

Receive working capital immediately. 70% held securely until delivery.



Disaster Protection

Submit proof-based aid claims for treasury compensation if crops fail.

FOR BUYERS & VALIDATORS

Confidence Before Commitment



There was an error generating this image

Commodity Buyers

Browse verified listings with NDVI health rings and AI crop summaries. Compare parcels objectively — not on a seller's word. Purchase with USDC knowing funds are protected by a Soroban escrow contract.

Validators & Insurers

Registered validators verify delivery outcomes — **Delivered, Disaster, or Fraud** — triggering automated escrow release or refund. Neither side can unilaterally walk away.

Everything You Need, On-Chain



Satellite-Verified Crop NFTs

Every listing gated by real Copernicus Sentinel-2 NDVI data.



70/20/10 Escrow Split

70% held until delivery, 20% upfront to farmer, 10% to aid pool.



Interactive Parcel Map

Draw polygons, bookmark parcels, run satellite checks on Google Maps.



AI Crop Summaries

NVIDIA Llama 3.3 translates satellite data into plain-language analysis.



Farmer ID Verification

Government ID upload with on-chain profile attestation.



Wallet-Agnostic

Connect via Freightier, Albedo, Lobstr, or WalletConnect.

Three Soroban Contracts. One Protocol.

1

Crop NFT Contract

Mints crop-backed NFTs with full metadata — crop type, quantity, price, parcel bounding box, NDVI threshold. Manages lifecycle: Available → Reserved → Growing → Verified → Completed → Failed.

2

Escrow Contract

Holds USDC and enforces the 70/20/10 split atomically. `release_escrow` and `refund_after_disaster` callable only after Verification contract records a delivery decision.

3

Verification Contract

Records satellite NDVI attestations via oracle. Manages validator set, delivery decisions (Delivered/Disaster/Fraud), and cross-contract calls to unlock escrow.

Built for Reliability at Every Layer



There was an error generating this image

Why Stellar?

Sub-second finality, sub-cent fees, and native USDC support make Stellar ideal for agricultural payments where margins matter. Soroban provides deterministic escrow logic without gas unpredictability.

Smart Contracts

Rust · Soroban SDK 25.0.1 · Stellar
Testnet

Frontend

Next.js 16 · React 19 · Tailwind CSS v4 ·
GSAP

Satellite & AI

Copernicus Sentinel-2 via openEO ·
NVIDIA Llama 3.3 70B

What's Next for Agri-Block

1

Phase 1 — Live

Testnet deployed. Marketplace, satellite verification, and escrow operational on Stellar Testnet.

2

Phase 2 — Enhanced

Multi-chain satellite sources (Landsat, PlanetScope), automated NDVI alerts, mobile-first offline parcel drawing.

3

Phase 3 — Predictive

Harvest prediction models from multi-year NDVI time series. DAO-governed validator selection.

4

Phase 4 — Mainnet

Real USDC from regulated issuers. Full production deployment serving smallholder farmers globally.



Live Demo: agri-con.vercel.app · **License:** MIT